# Preliminaries

## Lecture 1

Termeh Shafie

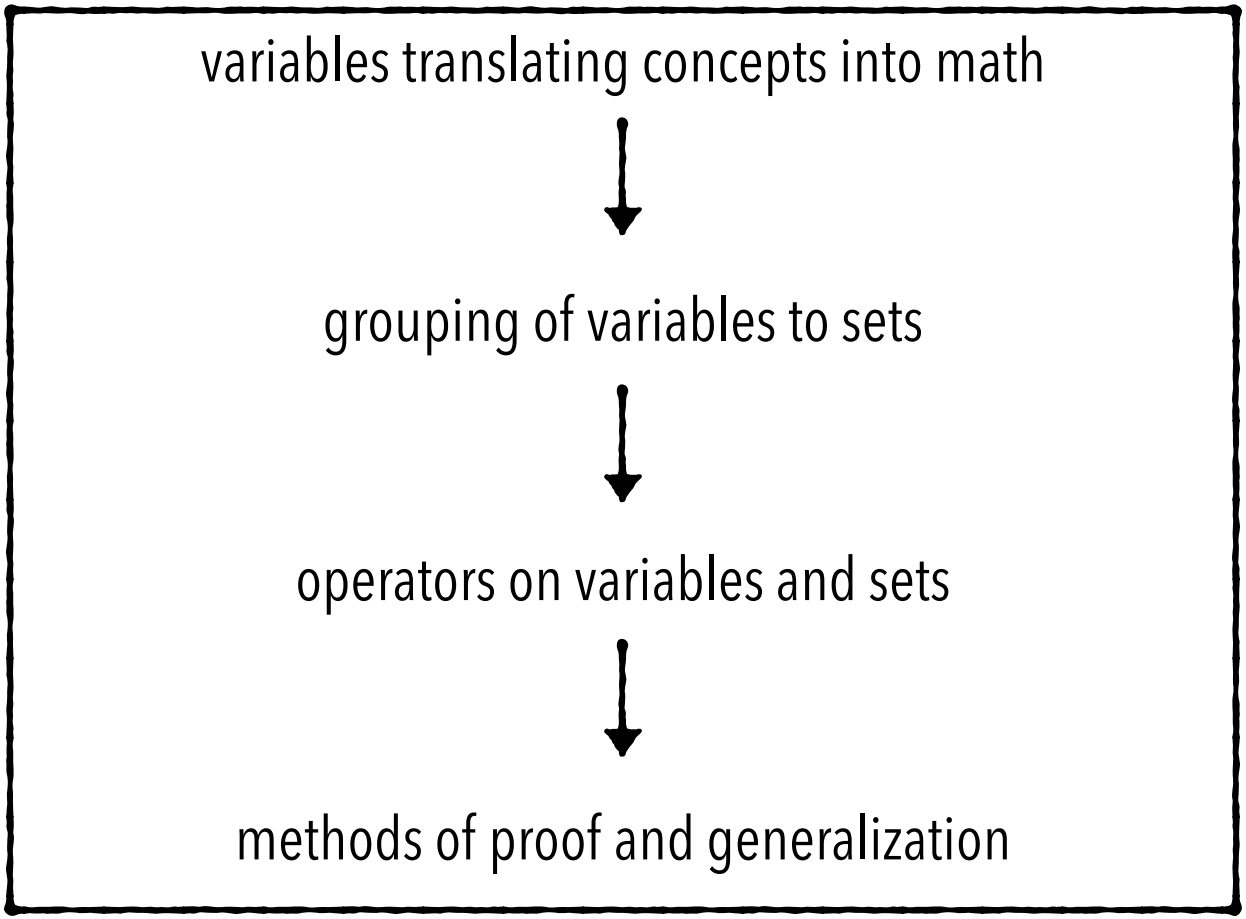

Welcome!
Everything is fine.

# roadmap of the course

Building blocks

I. Calculus in one dimension
II. Probability theory
III. Linear algebra
IV. Multivariate calculus and optimization

# preliminaries



# building a toy model of the world in math

**Theory**

- a set of statements involving concepts and concern relationships among abstract concepts

**Statements**

- comprise assumptions, propositions, corollaries, and hypotheses

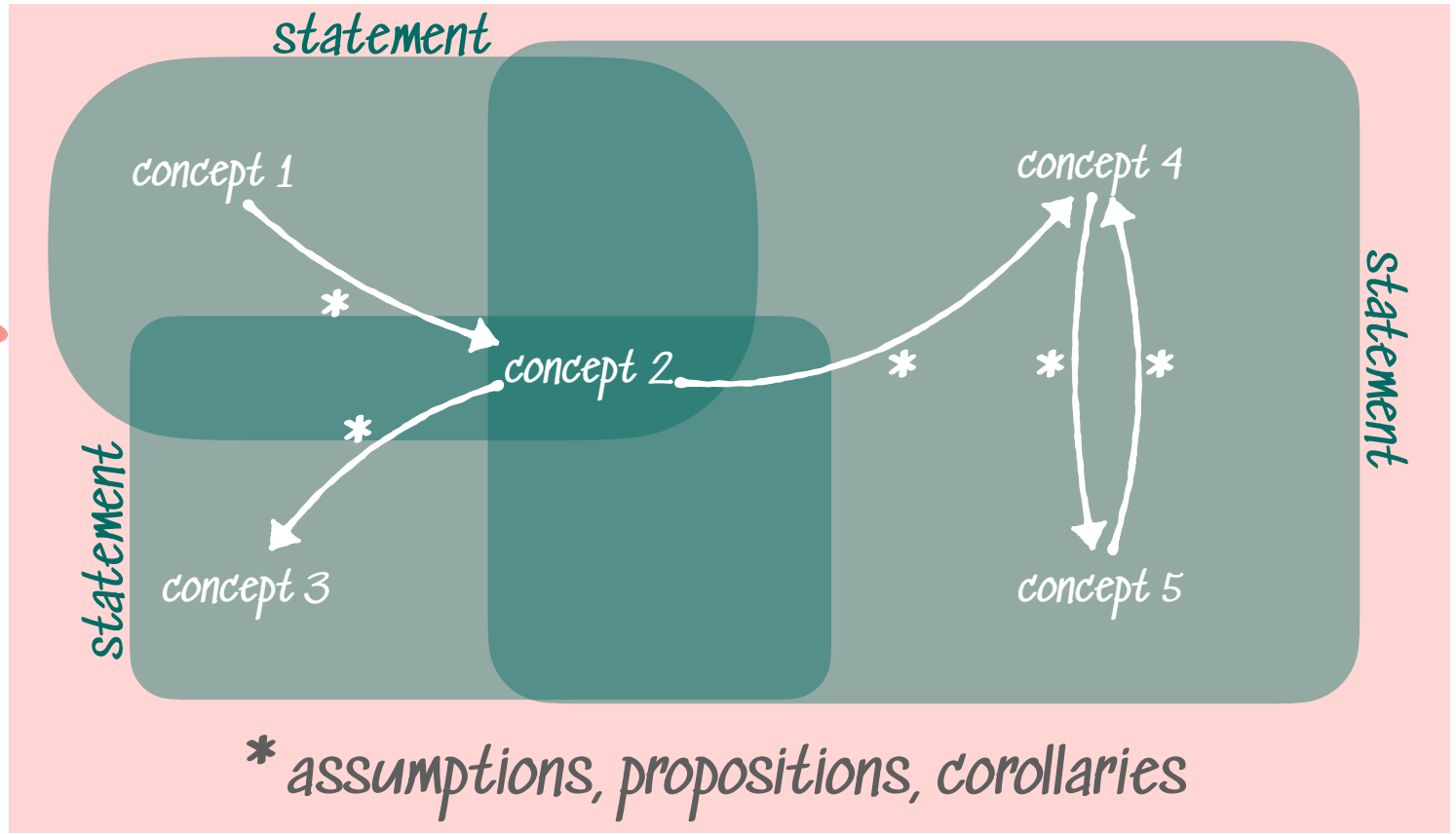
**Assumptions** are asserted by us

- propositions and corollaries are deduced from these assumptions
- **hypotheses** are derived from these deductions and then empirically challenged

**Concepts** help understand the world and can be operationalized into mathematical expressions with

- **variables**
  - are indicators we develop to measure our concepts
  - take on different values in a given set (i.e. it can vary)
- **constants:** take only one value for a given set (i.e. cannot vary)

# building a toy model of the world in math




⇑ hypotheses

# what is a set?

A set is a collection of elements or members

- curly braces { } used to list elements separated by comma ("Roster Method")
- Ellipsis (…) used within the braces to indicate that list continues in established pattern
- Cardinality of a set: the number of distinct elements in a set

**example**

| | |
|---|---|
| set A: | the natural numbers from 1 to 7 |
| elements of A: | 1,2,3,4,5,6,7 |
| set notation: | $A = \{1,2,3,4,5,6,7\} = \{1,2,3,\ldots,7\}$ |
| cardinality: | $\lvert A \rvert = 7$ |

# what is a set?

Russell's Paradox

Suppose a town's barber shaves all men in the town,

and only those men, who do not shave themselves.

***Who shaves the Barber?***

Consider the set S of all sets which do not contain themselves. Does S contain itself?



sets describe variables as discrete or continuous

- a variable is discrete if each one of its possible values are countable and can be matched one-to-one with a subset of the integers.
- a variable is continuous if its values cannot be assigned a single integer

➡ it can take any value in an interval of real numbers

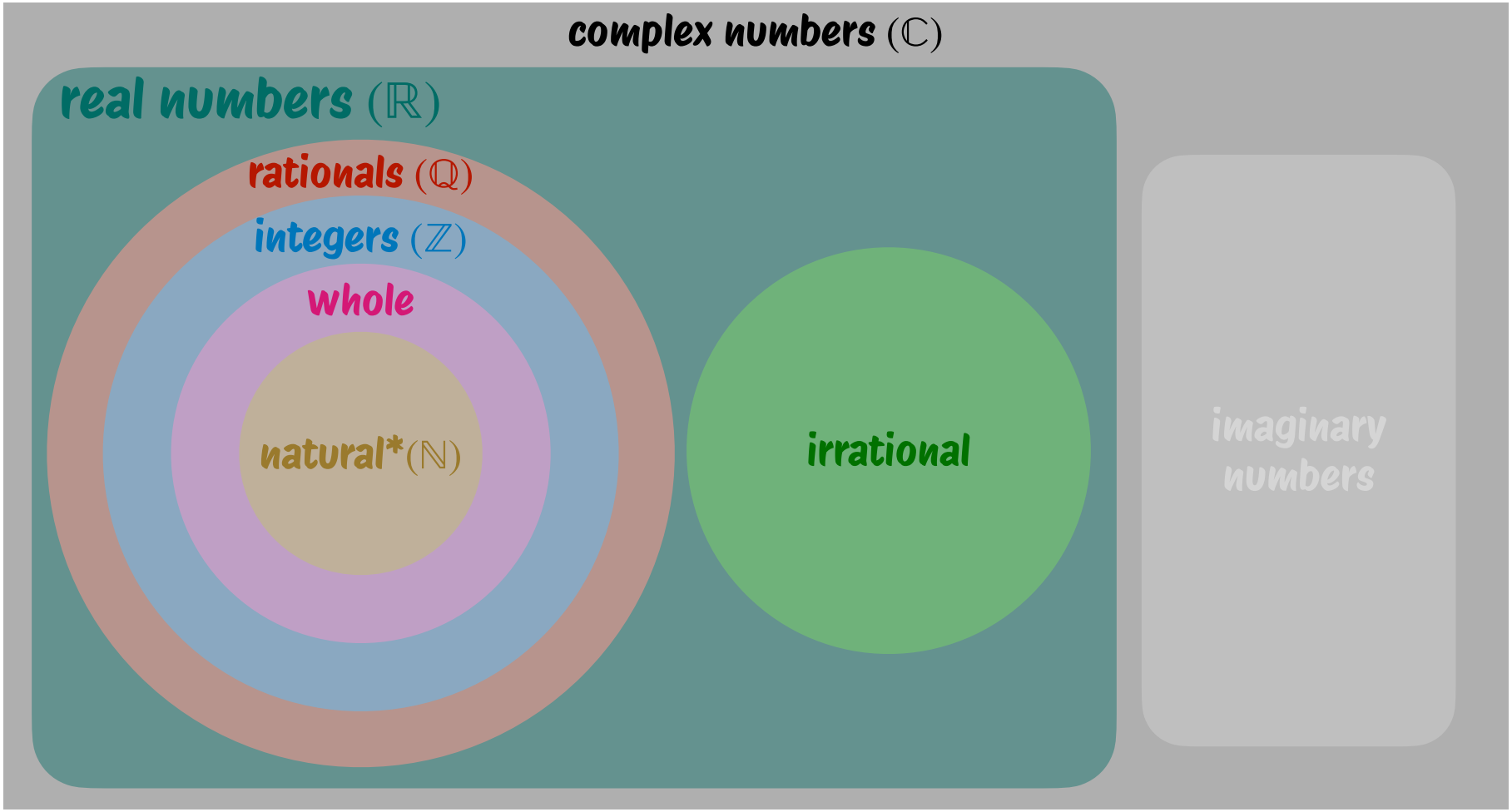
# set notation

- To say an element belongs to a set we use a "funky E": $\in$
- $A \subseteq B$ or $B \supseteq A$ means set A is a subset of set B
- $A \subset B$ means that A is a proper subset of B

# more on this in your tutorial:

- Finite/Infinite
- Countable/Uncountable
- Bounded/Unbounded
- Singleton
- Tuple
- Empty
- Universal
- Ordered/Unordered
- Solution set
- Sample space

# common sets




$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$

*some textbooks include 0 in natural numbers, denoted $\mathbb{N}_0$, so always check the convention.

## basic operators

- addition $+$
- subtraction $-$
- multiplication $\times$
- division $\div$
- exponentiation $x^a$
- *n*th root $\sqrt[n]{x}$
- factorial $!$
- sum $\sum_i x_i$
- product $\prod_i x_i$

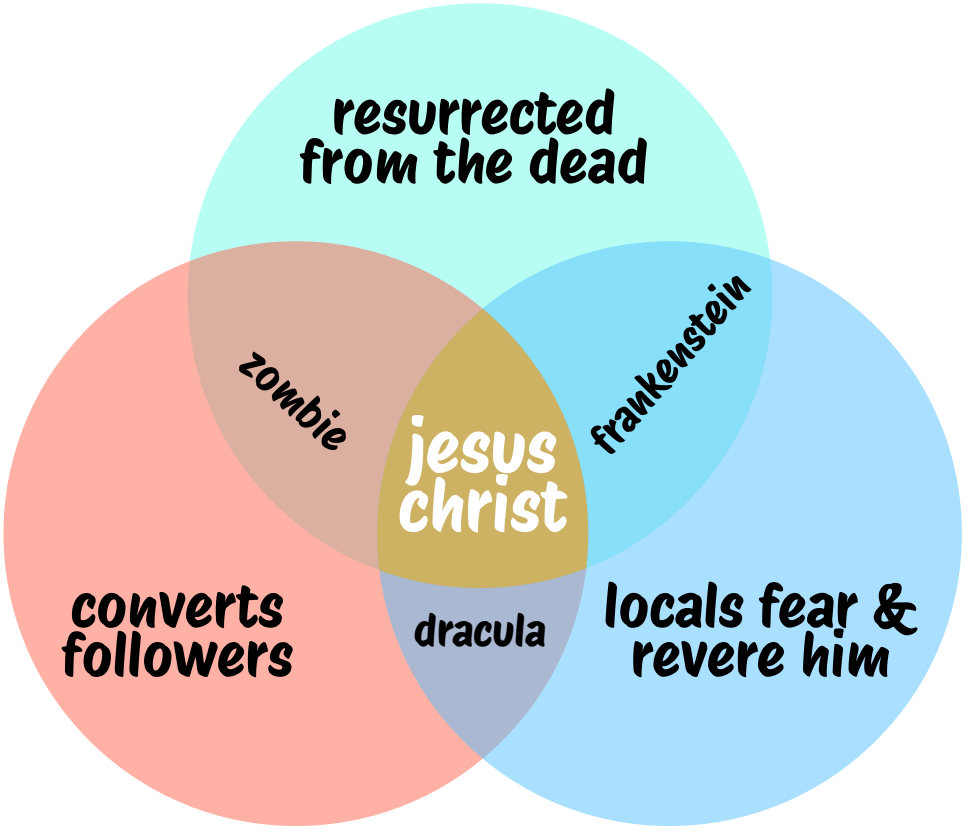
## set operators

- difference $A \setminus B$ or $A - B$
- complement $A'$ or $A^c$ or $\bar{A}$ or $\neg A$
- intersection $A \cap B$
- union $A \cup B$
- mutually exclusive $A \cap B = \varnothing$
- Cartesian product.
  $A \times B = \{(a, b) \mid a \in A, b \in B\}$
- symmetric difference
  $A \oplus B = (A - B) \cup (B - A)$
- partition:
  collection of subsets whose union forms the set

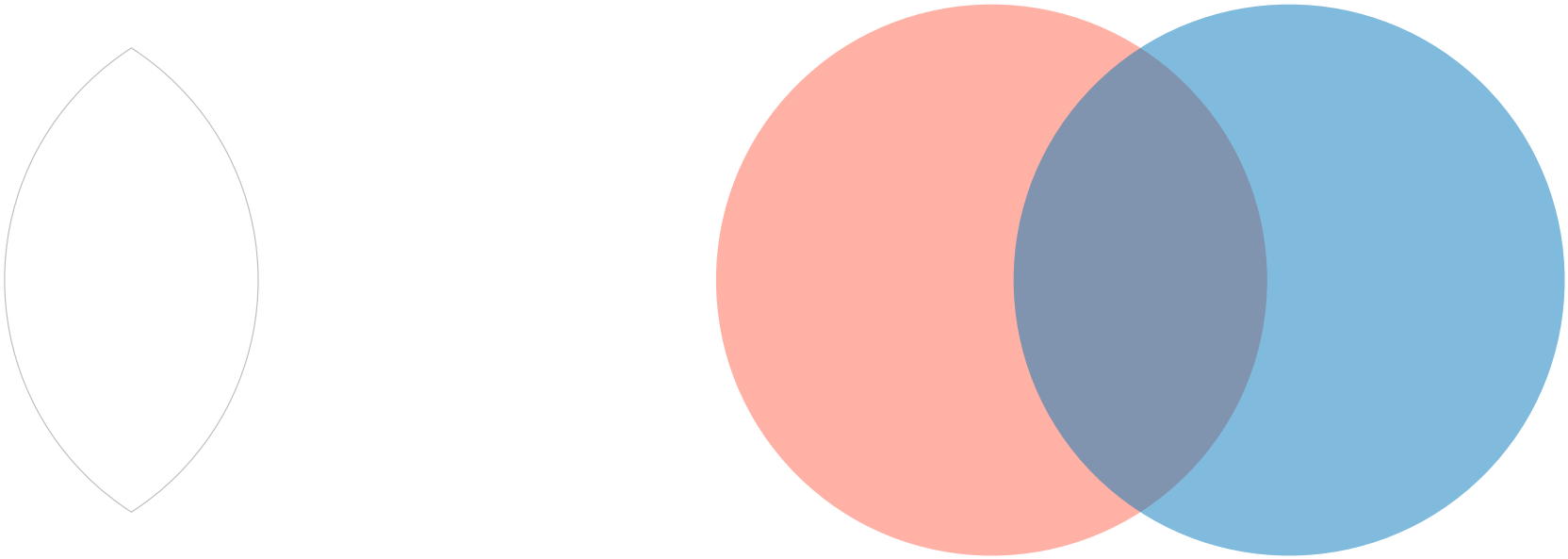
...more on this in your tutorial

# Venn diagrams

popular "thanks" to social media but often used incorrectly



# set operators with Venn diagrams

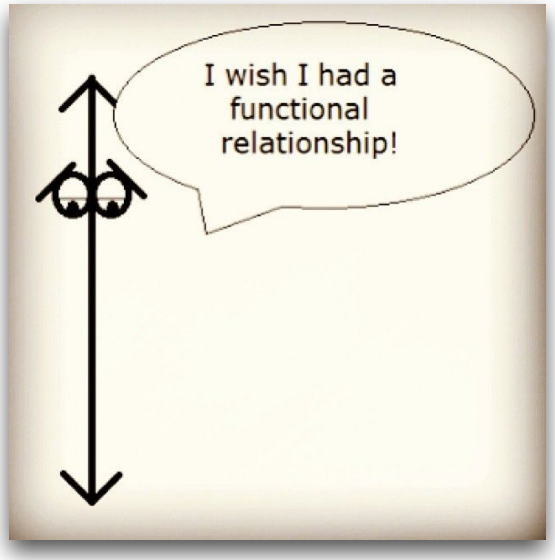


cardinality of the set union

$$|A \cup B| = |A| + |B| - |A \cap B|$$

# relations and functions

used to compare concepts and uncover relationships between them



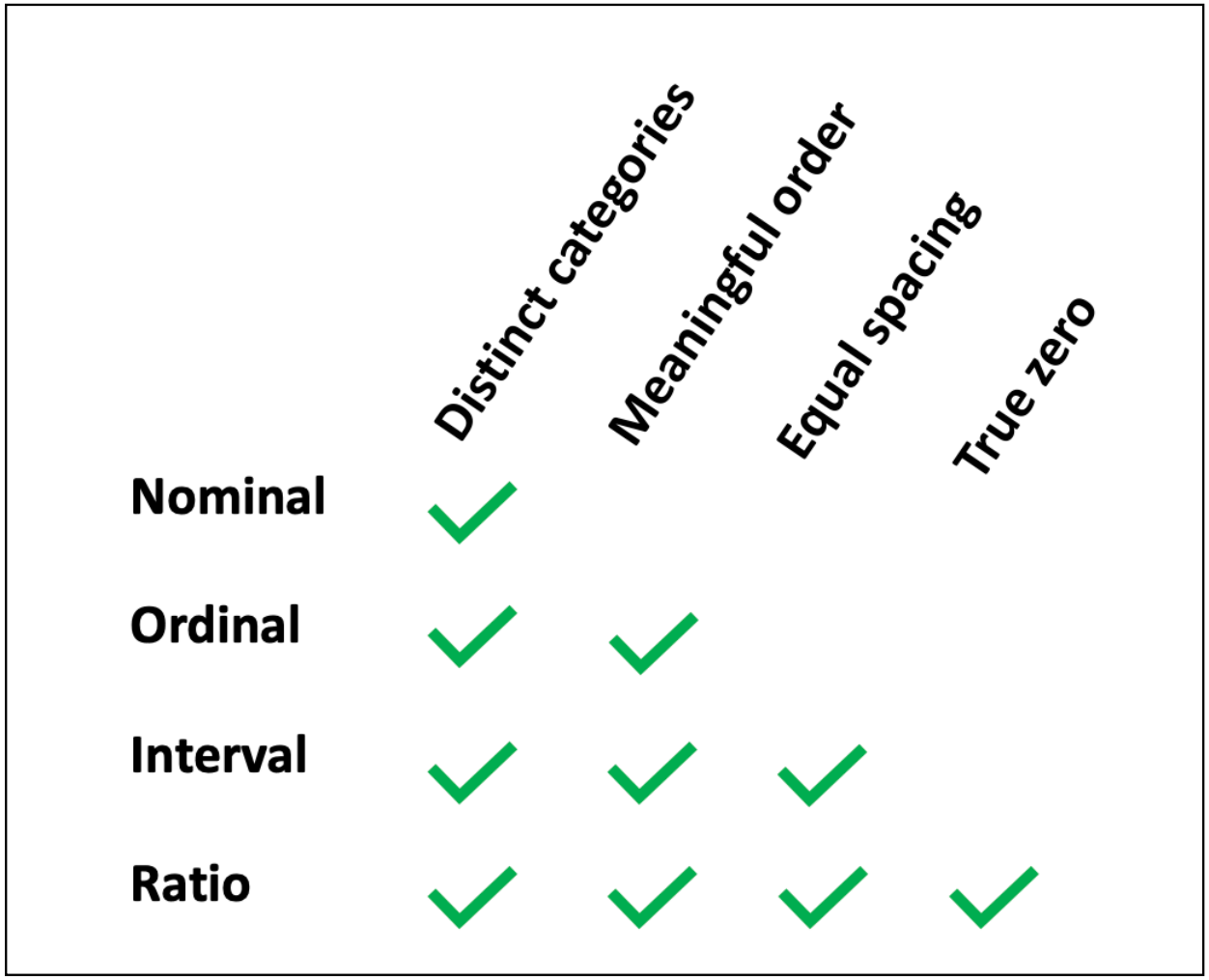

- a **relation** is a relationship between sets of information
- a **function** is a well-behaved relation

consider a function $f(x)$:

**domain** → The domain consists of all possible values that x can take on
**range** → The range consists of all possible values y takes on given x

*more on this in lecture 3...*

# level of measurement



# mathematical proofs

**Axioms and assumptions**

- stated to begin and assumed as true

**Proposition**

- a statement that is proven, or that is to be proven.

**Theorem**

- a proven proposition

**Lemma**

- a theorem used as a prior step to solve another problem (an auxiliary result)

**Corollary**

- a result that follows readily from a theorem, with little or no additional proof

# mathematical proofs

a **proof i**s an argument that demonstrates why a conclusion is true, subject to certain standards of truths

a **mathematical proof i**s an argument that demonstrates why a statement is true, following the rules of mathematics

## direct proofs

proof by deduction

proof by induction

proof by exhaustion

proof by construction

## indirect proofs

proof by contradiction

proof by contrapositive

# our first proof (by construction)

**Theorem**

For all integers $n$, if $n$ is even, then $n^2$ is even.




- Find the **formal definitions** for any terms in the theorem:
  - an integer $n$ is called **even** if there is an integer $k$ where $n = 2k$
  - an integer $n$ is called **odd** if there is an integer $k$ where $n = 2k + 1$
- What is the grammatical structure of the theorem?
  - **For all** integers $n$, **if** $n$ is even, **then** $n^2$ is even.

# our first proof (by construction)

**Theorem**

For all integers $n$, if $n$ is even, then $n^2$ is even.




- Pick some arbitrary even integer $n$ and try some examples:
  - $2^2 = 4 = 2 \times 2$
  - $10^2 = 100 = 2 \times 50$
  - $0^2 = 0 = 2 \times 0$
  - $(-8)^2 = 64 = 2 \times 32$
  - $n^2 = = 2 \times ?$

what's the pattern?
can we predict this?

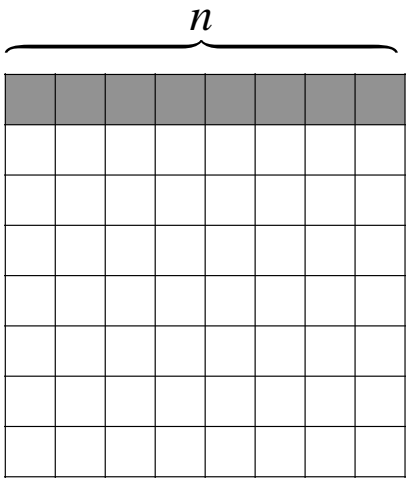
# our first proof (by construction)

**Theorem**

For all integers $n$, if $n$ is even, then $n^2$ is even.




- If possible, it's helpful to draw some pics




➡ an integer $n$ is called **even** if there is an integer $k$ where $n = 2k$

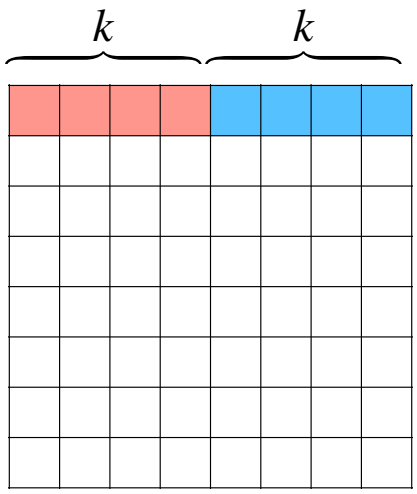
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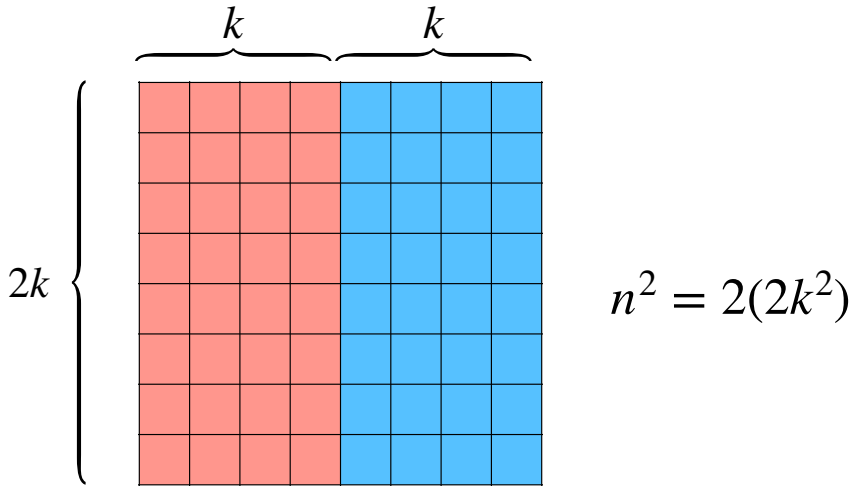
# our first proof (by construction)

**Theorem**

For all integers $n$, if $n$ is even, then $n^2$ is even.




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$$n^2 = 2(2k^2)$$

# our first proof (by construction)

**Theorem**

For all integers $n$, if $n$ is even, then $n^2$ is even.




**Proof.**

- Pick an arbitrary even integer $n$: we want to show that $n^2$ is even
- Since $n$ is even, there is some integer such that $n = 2k$
- This means that $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$
- From this we see that there is an integer $m$ (namely $2k^2$) where $n^2 = 2m$
- Therefore $n^2$ is even, which is what we wanted to show. ■

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end of proof "drop the mic"



# let's try another

**Theorem**

For all integers $m$ and $n$, if $m$ and $n$ are odd, then $m+n$ is even.




- Find the **formal definitions** for any terms in the theorem:
  - an integer $n$ is called **even** if there is an integer $k$ where $n = 2k$
  - an integer $n$ is called **odd** if there is an integer $k$ where $n = 2k + 1$
- What is the grammatical structure of the theorem?
  - **For all** integers $m$ and $n$, **if** $m$ and $n$ are odd, **then** $m+n$ is even.

# let's try another

**Theorem**

For all integers $m$ and $n$, if $m$ and $n$ are odd, then $m+n$ is even.




- Visual intuition





➡ an integer $n$ is called **odd** if there is an integer $k$ where $n = 2k + 1$

# let's try another

**Theorem**

For all integers $m$ and $n$, if $m$ and $n$ are odd, then $m+n$ is even.




- Visual intuition

$2k+1$ 

 $2r+1$

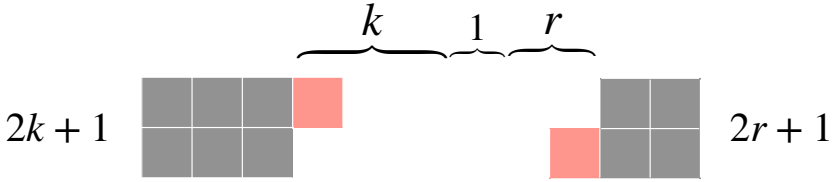
# let's try another

**Theorem**

For all integers $m$ and $n$, if $m$ and $n$ are odd, then $m+n$ is even.




- Visual intuition



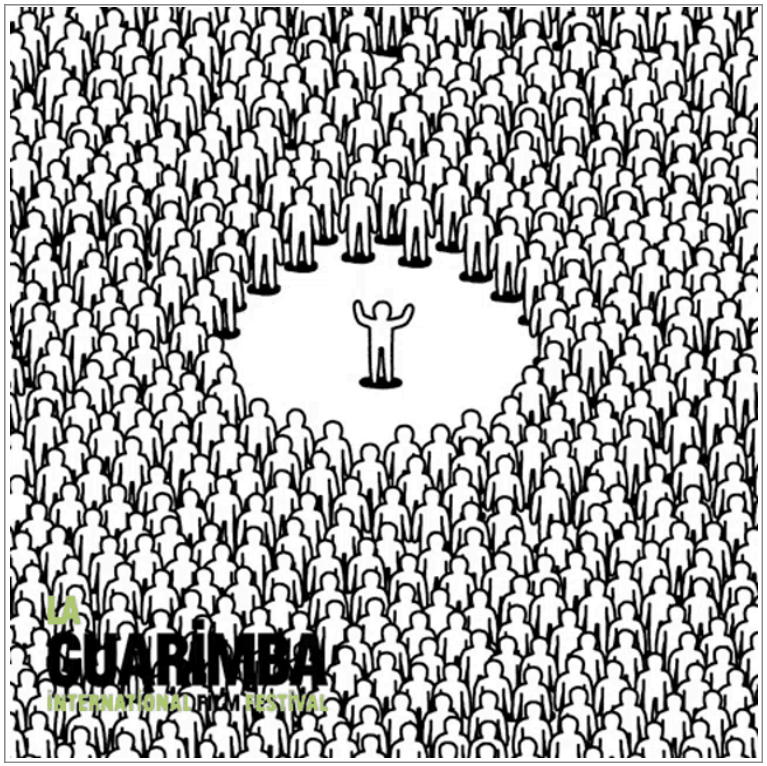

$$(2k+1)+(2r+1)=2(k+r+1)$$
$$m \quad + \quad n \quad = 2( \quad s \quad )$$

➡ an integer $n$ is called **even** if there is an integer $k$ where $n = 2k$

exercise: finish writing this proof by yourself

# the principle of mathematical induction

**everybody do the wave!**



# the principle of mathematical induction

let $P$ be some predicate

If $P(0)$ is true and $\forall k \in \mathbb{N}_0 \; P(k) \rightarrow P(k+1)$, then $\forall n \in \mathbb{N}_0 \; P(n)$

**if it starts true** **and it stays true** **then it's always true**

- it is true for 0
- since it's true for 0, it's true for 1
- since it's true for 1, it's true for 2
- since it's true for 2, it's true for 3
- since it's true for 3, it's true for 4

⋮

# proof by induction

- use the principle of mathematical induction to show that some result is true for all natural numbers $n$
- the proof, step by step:
  1. **<u>The base case</u>:** prove that $P(0)$ is true
  2. **<u>Inductive step:</u>** prove that if $P(k)$ is true then $P(k+1)$ is true
  3. Conclude by induction that $P(n)$ is true for all $n \in \mathbb{N}_0$



# proof by induction

**Theorem**

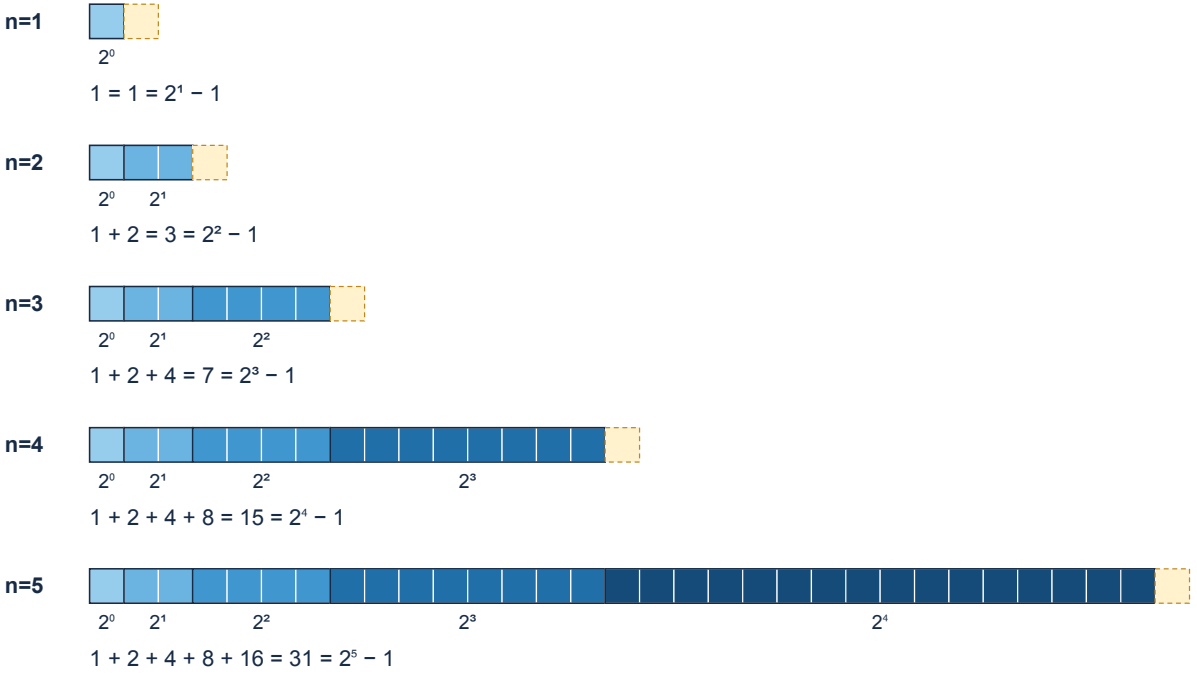
The sum of the first $n$ powers of two is $2^n - 1$.




## Proof visual:

$2^0 + 2^1 + \cdots + 2^{n-1} = 2^n - 1$

Each square = 1 unit. The dashed square is the one missing unit.



# proof by induction

**Theorem**

The sum of the first $n$ powers of two is $2^n - 1$.




**Proof.**

- Let $P(n)$ be the statement "the sum of the first $n$ powers of two is $2^n - 1$."
- We prove by induction, that $P(n)$ is true for all $n \in \mathbb{N}$ from which the theorem follows
- The base case:
  - we need to show $P(1)$ is true, meaning that the sum of the first one power of two is $2^1 - 1$.
  - since the sum of the first one power of two is $2^0 = 1$, and $2^1 - 1 = 1$, we see that $P(1)$ is true. ✓

# proof by induction

**Theorem**

The sum of the first $n$ powers of two is $2^n - 1$.




## Proof cont'd.

- The inductive step:
  - the goal here is to prove "if $P(k)$ then $P(k+1)$ is true"
  - to do this we choose an arbitrary $k$, assume $P(k)$ is true, then try to prove $P(k+1)$

    $\Longrightarrow$ assume that for some arbitrary $k \in \mathbb{N}$ that $P(k)$ holds, meaning

    $$2^0 + 2^1 + \cdots + 2^{k-1} = 2^k - 1$$
  - we need to show that $P(k+1)$ holds, meaning the sum of the first $k+1$ powers of two is $2^{k+1} - 1$

    $$\begin{aligned} 2^0 + 2^1 + \cdots + 2^{k-1} + 2^k &= 2^k - 1 + 2^k \\ &= 2(2^k) - 1 \\ &= 2^{k+1} - 1 \end{aligned}$$

    ✓
- Therefore $P(k+1)$ is true, so by induction $P(n)$ holds for all n $\in \mathbb{N}$. ■

# indirect proofs

- Proof by contrapositive

to prove the statement
***"if P is true, then Q is true"***
you instead prove the equivalent statement
***"if Q is false, then P is false"***

- Proof by contradiction

to prove the statement
***"if P is true, then Q is true"***
you show that the following is not possible
***"assume P is true and Q is false, and derive a contradiction"***

- Proof by counterexample (a universal "for all" claim is false)

# indirect proofs: proof by contrapositive

**Theorem**

For any $n \in \mathbb{Z}$, if $n^2$ is even, then $n$ is even.




**Proof.**

- By contrapositive; we prove that if $n$ is odd, then $n^2$ is odd
- Let $n$ be an arbitrary odd integer.
- Since $n$ is odd, there is some integer $k$ such that $n = 2k + 1$.
- Squaring both sides of this equality and simplifying yields the following:

$$\begin{aligned} n^2 &= (2k+1)^2 \\ &= 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1 \end{aligned}$$

- From this we see that there is an integer $m$ (namely $2k^2 + 2k$) such that $n^2 = 2m + 1$.
- Therefore $n^2$ is odd. ■

# indirect proofs: proof by contradiction

**Theorem**

For any $n \in \mathbb{Z}$, if $n^2$ is even, then $n$ is even.




**Proof.**

- Assume for the sake of contradiction that $n$ is an integer and that $n^2$ is even, but that $n$ is odd.
- Since $n$ is odd, there is some integer $k$ such that $n = 2k + 1$.
- Squaring both sides of this equality and simplifying yields the following:

$$\begin{aligned} n^2 &= (2k+1)^2 \\ &= 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1 \end{aligned}$$

- This tells us that $n^2$ is odd, which is impossible, by assumption $n^2$ is even.
- We have a contradiction so our assumption is incorrect
  $\Longrightarrow$ if $n$ is an integer and $n^2$ is even then $n$ is also even. ■